Food Redistribution

Proof of Concept Report

Detail	Information
Project Title	Food Redistribution
Team Members	Balaji V - 2666356 Pavithra S - 2665966 Dhanush A S - 2666824
Submitted By	Team Trinity
Submission Date	27 March 2026

1. INTRODUCTION

Food wastage is a growing concern across the world, especially in countries like India where large-scale celebrations such as weddings, corporate events, and social gatherings are a regular part of life. Ironically, while a significant portion of food gets discarded after such events, there are still countless individuals and communities who struggle to access even a single nutritious meal a day. This imbalance highlights a critical gap—not in availability, but in distribution and timely coordination.

In most real-life scenarios, individuals or event organizers who wish to donate excess food often face practical challenges. They may not know which nearby shelters or organizations can accept food donations, or they may spend valuable time searching online and making multiple calls. By the time they identify a suitable recipient, the food may no longer be in a condition suitable for consumption. As a result, despite good intentions, much of the surplus food ends up being wasted.

To address this issue, we envisioned a solution that simplifies and accelerates the entire donation process. The idea was to create a system that requires minimal effort from the donor, eliminates the need for manual searching, and ensures quick connection with the right shelter. Keeping accessibility in mind, we designed the solution to work through a simple phone call, making it usable even for people who are not comfortable with mobile applications or digital platforms.

This project presents a Proof of Concept (POC) for a Food Redistribution System that leverages cloud technologies and voice-based interaction to bridge the gap between food donors and shelters. By integrating services such as automated voice assistance, real-time location tracking, and workflow orchestration, the system enables seamless communication and efficient matching between donors and recipients.

The goal of this solution is not only to reduce food wastage but also to demonstrate how technology can be used to create meaningful social impact. By enabling faster decision-making, reducing manual effort, and ensuring reliable coordination, this system lays the foundation for a scalable platform that can contribute towards a more sustainable and equitable food distribution ecosystem.

2. EXECUTIVE SUMMARY

Overview of the Solution

The Food Redistribution System is a cloud-based, voice-enabled solution designed to simplify and accelerate the process of donating surplus food. The system allows donors to initiate a donation request through a simple phone call, eliminating the need for searching or navigating complex applications. Using intelligent voice interaction and automated workflows, the system captures key details such as food quantity, location, and type, and then connects the donor with the most suitable nearby shelter.

Built entirely on a serverless architecture using AWS services and integrated with Twilio for communication, the solution ensures scalability, reliability, and minimal operational overhead. The system not only identifies nearby shelters based on geographic proximity but also automates outreach and confirmation, creating a seamless end-to-end donation experience.

Problem Summary

Food wastage and hunger coexist as parallel challenges, primarily due to inefficiencies in distribution and lack of real-time coordination.

- Excess food is frequently wasted after events and gatherings
- Donors struggle to find nearby shelters quickly
- Time delays reduce food usability and quality
- Manual coordination is inefficient and unreliable
- Dependence on internet/apps limits accessibility

These issues often discourage donors from taking action, resulting in missed opportunities to redistribute food effectively.

Solution Summary

The proposed solution addresses these challenges through automation, voice interaction, and intelligent location-based matching.

- **Voice-based donation system** – Donors simply make a phone call

- **Conversational AI (Amazon Lex)** captures donation details
- **Location intelligence** identifies nearby shelters dynamically
- **Automated outreach system** contacts shelters sequentially
- **OTP-based confirmation** ensures secure and verified transactions
- **Serverless backend** ensures scalability and low maintenance

By removing friction from the process, the system enables faster and more efficient food redistribution.

User Impact

This solution is designed to create meaningful impact for all stakeholders involved:

- **For Donors**
    - No need to search manually for shelters
    - Quick and effortless donation process
    - Increased likelihood of food being utilized
- **For Shelters**
    - Real-time notifications of available food
    - Ability to accept or decline based on capacity
    - Reduced dependency on manual outreach
- **For Society**
    - Reduction in food wastage
    - Better support for underprivileged communities
    - Encouragement of responsible consumption practices

Innovation Summary

The uniqueness of this solution lies in its simplicity, accessibility, and intelligent automation.

- **Voice-first approach** eliminates dependency on smartphones or apps
- **End-to-end automation** from request capture to fulfillment
- **Real-time location-based matching** using geospatial services
- **Dynamic retry mechanism** ensures maximum success rate
- **Fully serverless architecture** for cost efficiency and scalability
- **Bridges the gap between intention and action** in food donation

3. PROBLEM STATEMENT

Real-World Challenge

Food wastage is a significant global issue, particularly in urban environments where large-scale events, celebrations, and gatherings are frequent. In such settings, a considerable amount of surplus food is generated, much of which is discarded despite being safe and suitable for consumption. At the same time, there are communities and individuals who continue to face food insecurity on a daily basis.

The core challenge is not the lack of food availability, but the absence of an efficient and timely mechanism to redistribute surplus food. Donors who are willing to contribute often encounter difficulties in identifying nearby shelters or organizations that can accept food at short notice. The process of searching, contacting, and coordinating with shelters is typically manual, time-consuming, and unreliable. As a result, the window of opportunity to redistribute food is often missed, leading to unnecessary wastage.

Who is Affected

The problem impacts multiple stakeholders across the ecosystem:

- **Event organizers and donors**
  Individuals and organizations hosting events frequently have surplus food but lack a quick and reliable way to donate it. This leads to frustration and, in many cases, food being discarded despite the willingness to donate.
- **Shelters and non-profit organizations**
  Many shelters depend on external food donations to support their operations. However, they often do not receive timely information about available food, limiting their ability to respond and utilize these resources effectively.
- **Underprivileged communities**
  The ultimate impact is felt by individuals and families who rely on shelters for meals. Missed opportunities in redistribution directly translate to missed meals for those in need.
- **Society and environment**
  Food wastage contributes to environmental issues, including increased landfill use

and greenhouse gas emissions. Inefficient redistribution systems exacerbate these problems.

Evidence and Examples

The issue of food wastage alongside food scarcity is well-documented and observable in everyday scenarios:

- At large events such as weddings or corporate functions, excess food is often disposed of due to the absence of immediate redistribution options.
- Donors typically rely on internet searches or personal contacts to find nearby shelters, which can take significant time and may not always yield reliable results.
- In many cases, by the time a suitable shelter is identified and contacted, the food may no longer be in a condition suitable for donation.
- Shelters, on the other hand, may have the capacity to accept food but remain unaware of available donations due to the lack of a real-time communication mechanism.

These examples highlight a clear gap between food surplus and food demand, driven primarily by inefficiencies in discovery, communication, and coordination. Addressing this gap requires a solution that is fast, accessible, and capable of operating in real time.

4. PROPOSED SOLUTION

Solution Description

The proposed solution is a cloud-based, voice-enabled Food Redistribution System designed to streamline and automate the process of donating surplus food. The system allows donors to initiate a donation request through a simple phone call, removing the need for manual searching or dependency on digital applications.

Once the call is initiated, the system leverages a conversational interface to collect essential details such as the quantity of food, location (via zip code), and type of food. This information is processed through a serverless backend that orchestrates the entire workflow, including identifying suitable shelters, initiating communication, and confirming the donation.

The solution is built using a combination of cloud services that enable real-time processing, scalability, and reliability. It integrates voice communication, intelligent decision-making, and location-based services to create a seamless end-to-end experience for both donors and shelters.

How It Solves the Problem

This solution directly addresses the key challenges associated with food redistribution by introducing automation, speed, and accessibility into the process.

- **Eliminates manual search**
  Donors no longer need to spend time searching for shelters. The system automatically identifies nearby shelters based on the provided location.
- **Reduces time delays**
  By capturing information instantly through a phone call and processing it in real time, the system ensures that food is matched with a recipient before it becomes unusable.
- **Enables real-time coordination**
  Automated calls are placed to multiple shelters in sequence, allowing quick acceptance or rejection without manual intervention.

- **Improves accessibility**

  The voice-based interface ensures that the system can be used by anyone with access to a phone, regardless of technical proficiency or internet availability.
- **Ensures reliability and confirmation**

  OTP-based verification provides a secure and reliable way to confirm the completion of the donation process for both parties.

Unique Value and Innovation

The solution introduces several innovative aspects that differentiate it from traditional approaches to food donation and redistribution:

- **Voice-first interaction model**

  Unlike typical platforms that rely on mobile or web applications, this system operates entirely through a phone call, making it highly accessible and user-friendly.
- **End-to-end automation**

  The entire process—from capturing donor intent to confirming shelter acceptance—is automated, reducing human dependency and minimizing delays.
- **Dynamic location-based matching**

  The system intelligently identifies shelters within a defined radius and dynamically expands the search area if no immediate match is found, increasing the chances of successful redistribution.
- **Sequential outreach mechanism**

  Instead of relying on a single point of contact, the system systematically reaches out to multiple shelters until a match is found, ensuring higher efficiency.
- **Serverless and scalable architecture**

  Built using cloud-native services, the solution is designed to scale seamlessly based on demand while maintaining low operational overhead.
- **Bridging intent and action**

  One of the most significant innovations is the ability to convert a donor's intention into immediate action with minimal effort, thereby reducing the gap between willingness to donate and actual donation.

5. USE CASES

Use Case Title: Food Donation and Redistribution via Voice-Based System

Actors

- Donor
  Individual or organization willing to donate surplus food.
- System
  The integrated platform consists of Twilio, Amazon Lex, AWS services (Lambda, Step Functions, Location Service, API Gateway), and DynamoDB.
- Shelter/Receiver
  Registered shelter or organization capable of accepting food donations.

Preconditions

- The donor has access to a phone and calls the designated service number.
- The system is operational with all cloud services properly configured.
- Shelter data (including contact details and location) is available in the database.
- Network connectivity is available for voice interaction and backend processing.
- The donor has valid and accurate information regarding food quantity and location.

Workflow Steps

1. Call Initiation
   The donor initiates the process by calling the designated number powered by Twilio.
2. Intent Capture and Interaction
   The call is handled by the conversational bot, which identifies the donor's intent to donate food and collects necessary details:
   - Quantity of food (via keypad input)
   - Zip code (via dial pad input)
   - Type of food (via voice input)
3. Data Processing and Workflow Trigger
   The collected information is passed to the backend through API Gateway, triggering a Step Functions workflow.

4. Location-Based Shelter Identification

   The system uses the provided zip code to identify nearby shelters using location services:

   - Searches within an initial radius (e.g., 10 km)
   - Expands the radius dynamically if no shelters are found
   - Retrieves a list of nearest available shelters

5. Automated Shelter Outreach

   The system initiates calls to the identified shelters sequentially:

   - Provides details about the available food
   - Prompts the shelter to accept or reject the request (e.g., press 1 to accept, press 2 to reject)
   - Continues contacting shelters until one accepts

6. Acceptance and Notification

   Once a shelter accepts:

   - The donor receives shelter details
   - The shelter receives donor details
   - Both parties receive an OTP for verification

7. OTP Verification and Completion

   The donor and shelter verify the transaction using the OTP:

   - Upon successful verification, the donation process is marked as complete
   - All details are stored in the database for record-keeping

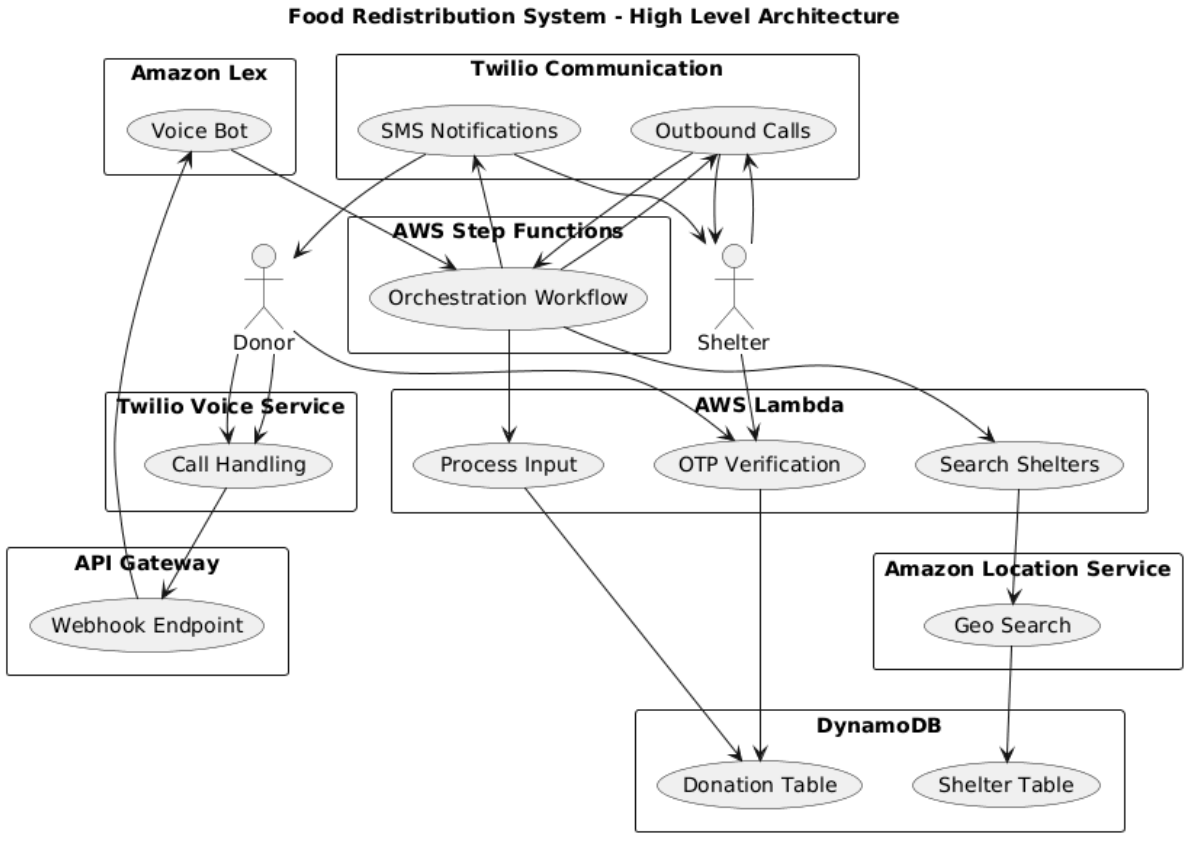
Success Criteria

- The donor is able to successfully initiate a donation request through a phone call.
- The system accurately captures and processes all required inputs.
- At least one suitable shelter is identified and contacted.
- A shelter accepts the donation request within the defined workflow.
- Both donor and shelter receive correct details and OTP verification.
- The donation is successfully confirmed and recorded in the system.
- The entire process is completed within a time frame that ensures the food remains usable.

6. ARCHITECTURE & DESIGN

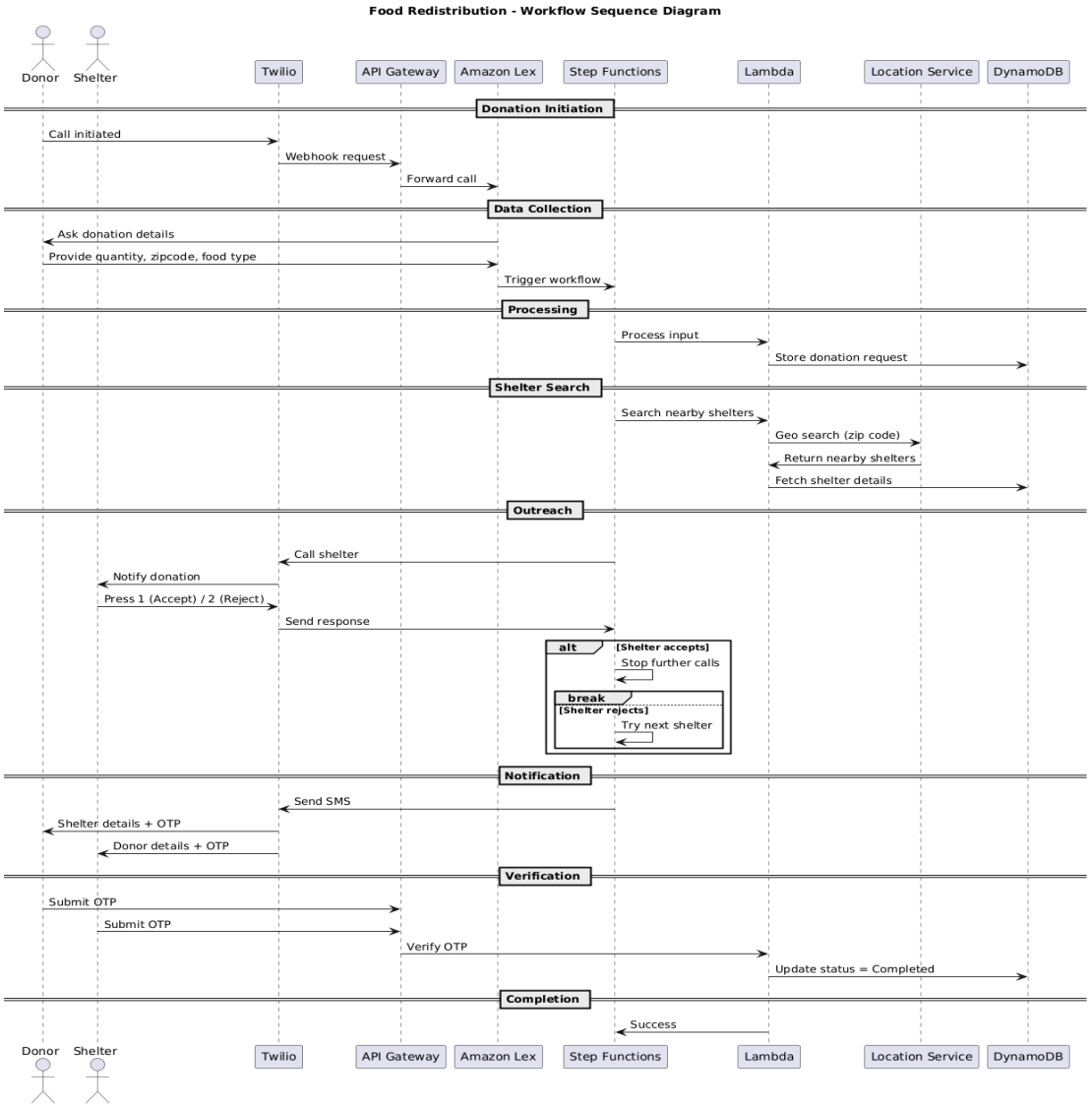
High-level architecture diagram

The architecture illustrates a serverless, event-driven system designed to enable seamless food donation and redistribution. The flow begins with the donor initiating a call through Twilio, which is routed via API Gateway to an Amazon Lex bot for conversational data capture. The collected inputs are processed through AWS Step Functions, which orchestrates multiple Lambda functions responsible for data handling, shelter discovery using Amazon Location Service, and communication workflows. DynamoDB is used for storing donation requests and shelter information, while Twilio handles outbound calls and SMS notifications. The architecture ensures scalability, reliability, and minimal operational overhead by leveraging fully managed cloud services.



Workflow Diagram

The workflow diagram represents the end-to-end interaction between the donor, system components, and shelters during the food donation process. It begins with the donor initiating a call, followed by conversational data collection through the Lex bot. The system processes this information and identifies nearby shelters based on location. It then sequentially contacts shelters until one accepts the donation request. Upon acceptance, both donor and shelter receive relevant details along with an OTP for verification. The process concludes with OTP validation and status update in the database, ensuring a secure and confirmed completion of the donation cycle. This workflow highlights real-time processing, automated decision-making, and efficient coordination between all participants.



Technology Stack Details

The proposed solution is built using a modern, cloud-native, serverless technology stack that ensures scalability, reliability, and minimal operational overhead. Each component in the system has been carefully selected to address specific functional requirements such as communication, orchestration, data processing, and storage.

Cloud Platform: Amazon Web Services (AWS)

The core infrastructure of the system is built on AWS, leveraging its serverless and managed services to ensure high availability and scalability without the need for infrastructure management. AWS enables rapid development, seamless integration between services, and cost optimization based on usage.

Communication Layer: Twilio

Twilio is used as the primary communication interface for both inbound and outbound interactions.

- Handles incoming calls from donors
- Integrates with backend systems using webhooks
- Enables automated outbound calls to shelters
- Sends SMS notifications for confirmation and OTP

The use of Twilio ensures that the system remains accessible via simple phone calls, removing dependency on internet-based applications.

API Management: Amazon API Gateway

API Gateway acts as the entry point for all external requests and integrations.

- Receives webhook requests from Twilio
- Routes requests to appropriate backend services
- Secures endpoints for OTP verification
- Ensures controlled and scalable API exposure

This enables seamless communication between external services and internal workflows.

Conversational Interface: Amazon Lex

Amazon Lex is used to build the voice-based conversational interface for interacting with donors.

- Captures user intent (e.g., food donation request)
- Collects structured inputs such as:
    - Quantity (via keypad input)
    - Zip code
    - Food type (via voice)
- Provides a natural and guided interaction experience

Lex enables efficient and accurate data collection without requiring a graphical interface.

Workflow Orchestration: AWS Step Functions

Step Functions coordinate the entire backend workflow in a structured and reliable manner.

- Manages the sequence of operations from input processing to final confirmation
- Handles conditional logic such as retrying multiple shelters
- Ensures fault tolerance and state management
- Improves visibility into the workflow execution

This allows the system to handle complex processes in a controlled and scalable way.

Compute Layer: AWS Lambda

AWS Lambda is used to execute backend logic in a serverless manner.

- Processes donor input data
- Implements business logic for:
    - Searching shelters
    - Managing outreach flow
    - OTP generation and verification
- Interacts with other AWS services (Location Service, DynamoDB)

Lambda ensures efficient execution without the need to manage servers, scaling automatically based on demand.

Location Intelligence: Amazon Location Service

Amazon Location Service is used to perform geospatial operations.

- Converts zip code into geographic coordinates
- Identifies nearby shelters within a defined radius
- Supports dynamic radius expansion if no shelters are found initially

This enables accurate and efficient matching of donors with nearby shelters.

Database: Amazon DynamoDB

DynamoDB is used as the primary data storage solution for the system.

- Stores donation requests and their statuses
- Maintains shelter information (location, contact details)
- Supports fast and scalable read/write operations
- Ensures high availability and low latency

Its serverless nature aligns with the overall architecture of the system.

Security and Verification: OTP Mechanism

An OTP-based verification mechanism is implemented to ensure secure and reliable confirmation of the donation process.

- OTP is generated and shared with both donor and shelter
- Verification is performed via secure API endpoints
- Ensures authenticity and prevents misuse

This adds a layer of trust and accountability to the system.

Architecture Characteristics

The overall technology stack is designed with the following key characteristics:

- **Serverless Architecture** – Eliminates infrastructure management and enables auto-scaling
- **Event-Driven Design** – Ensures real-time processing and responsiveness

- **High Availability** – Leveraging managed services ensures minimal downtime
- **Cost Efficiency** – Pay-as-you-go model reduces unnecessary expenses
- **Scalability** – Can handle increasing number of users and requests seamlessly

7. FEATURES & CAPABILITIES

1. Voice-Based Donation Initiation

Feature Description

The system allows donors to initiate the food donation process through a simple phone call. A conversational interface guides the user to provide necessary details such as food quantity, location, and type without requiring any application or internet access.

Business Value

- Increases accessibility for a wider audience, including non-technical users
- Encourages more participation in food donation due to ease of use
- Reduces dependency on digital platforms and manual coordination

Technical Value

- Utilizes conversational AI for structured data capture
- Integrates telephony services with cloud-based backend systems
- Ensures reliable input collection through voice and keypad interaction

2. Intelligent Location-Based Matching

Feature Description

The system identifies nearby shelters based on the donor's provided zip code. It performs geospatial searches within a defined radius and dynamically expands the search area if no immediate matches are found.

Business Value

- Ensures faster connection between donors and shelters
- Minimizes food spoilage by reducing response time
- Improves success rate of donations

Technical Value

- Implements geolocation services for accurate proximity search
- Supports dynamic radius expansion logic

- Enables efficient querying and filtering of location-based data

3. Automated Shelter Outreach

Feature Description

Once suitable shelters are identified, the system automatically contacts them via phone calls in a sequential manner. Each shelter is provided with donation details and can respond by accepting or rejecting the request.

Business Value

- Eliminates the need for manual calling and coordination
- Ensures timely outreach to multiple shelters
- Increases the likelihood of successful donation matching

Technical Value

- Implements event-driven communication workflows
- Supports retry and fallback mechanisms
- Integrates outbound communication APIs for automation

4. Real-Time Decision and Retry Mechanism

Feature Description

The system continues contacting shelters until one accepts the donation. If a shelter declines, the system automatically proceeds to the next available option without interrupting the workflow.

Business Value

- Ensures higher success rate in finding a recipient
- Reduces donor effort and waiting time
- Provides a reliable and consistent user experience

Technical Value

- Implements conditional logic and looping workflows
- Enables fault-tolerant orchestration

- Supports asynchronous processing and state management

5. OTP-Based Secure Verification

Feature Description

Once a shelter accepts the donation, both donor and shelter receive an OTP to verify and confirm the transaction. The donation is marked complete only after successful verification.

Business Value

- Builds trust between donors and shelters
- Prevents misuse or false confirmations
- Ensures accountability in the donation process

Technical Value

- Implements secure verification workflows
- Ensures data integrity and validation
- Integrates authentication mechanisms into the system

6. Automated Notifications

Feature Description

The system sends real-time notifications to both donor and shelter, including relevant details and confirmation messages once a match is found.

Business Value

- Keeps all stakeholders informed
- Reduces uncertainty and improves transparency
- Enhances overall user experience

Technical Value

- Supports real-time messaging through communication APIs
- Ensures reliable delivery of critical information
- Integrates notification services into workflow pipelines

7. Serverless and Scalable Architecture

Feature Description

The entire system is built using serverless cloud services, allowing it to scale automatically based on demand without manual infrastructure management.

Business Value

- Reduces operational and maintenance costs
- Supports large-scale adoption without performance issues
- Enables faster deployment and iteration

Technical Value

- Leverages managed cloud services for auto-scaling
- Ensures high availability and fault tolerance
- Supports modular and loosely coupled system design

8. Centralized Data Management

Feature Description

All donation requests, shelter details, and transaction statuses are stored in a centralized database, enabling efficient tracking and management.

Business Value

- Enables reporting and analytics for future improvements
- Provides visibility into system usage and impact
- Supports audit and tracking requirements

Technical Value

- Ensures low-latency data access
- Supports scalable and consistent data storage
- Enables integration with future analytics systems

8. INSTALLATION GUIDE

System Requirements

The system does not require any local installation or high-end hardware, as it is built entirely on cloud services. However, the following are required for setup and testing:

- An active AWS account with necessary permissions
- A Twilio account with a configured phone number
- Basic knowledge of AWS services and serverless architecture
- A web browser to access the OTP verification page
- Internet connectivity for configuration and testing

Backend Setup

The backend is implemented using AWS serverless services and does not require traditional deployment.

- Configure Amazon API Gateway to expose webhook endpoints for Twilio
- Create and deploy AWS Lambda functions for:
    - Processing donation requests
    - Shelter search logic
    - OTP generation and verification
- Set up AWS Step Functions to orchestrate the workflow
- Configure Amazon Lex bot with intents and slot types for capturing user inputs
- Integrate Twilio with API Gateway using webhooks for call handling and responses
- Ensure proper IAM roles and permissions are assigned to all services

Frontend Setup

The frontend component is minimal and consists of an OTP verification page.

- A simple HTML page is used for OTP input and verification
- The page connects to backend APIs (via API Gateway) for OTP validation
- Can be hosted using:
    - AWS S3 (static website hosting), or
    - Any lightweight web hosting service

No complex frontend framework or build process is required.

Database Setup

The system uses a fully managed NoSQL database.

- Create DynamoDB tables for:
    - Donation requests (status, quantity, location, etc.)
    - Shelter details (contact information, location data)
- Configure appropriate primary keys and indexes for efficient querying
- Ensure read/write permissions are granted to Lambda functions

Environment Variables

Environment variables are used to securely manage configuration values across services.

- Twilio credentials:
    - Account SID
    - Auth Token
    - Twilio phone number
- AWS configuration:
    - Region
    - Resource identifiers (Lambda, Step Functions, DynamoDB tables)
- Application-specific variables:
    - OTP expiry duration
    - Default search radius for shelter lookup

These variables are configured within Lambda functions or deployment settings to ensure secure and flexible operation.

9. USER MANUAL

Accessing the System

The system is designed to be easily accessible without requiring any login or application installation. Users can initiate the process by simply calling the designated service number provided through Twilio.

Providing Input

Once the call is connected, the user interacts with an automated voice assistant that guides them through the process. The system collects the following inputs:

- Donation intent (e.g., food donation request)
- Quantity of food (entered using keypad input)
- Location details such as zip code (entered via dial pad)
- Type of food (provided through voice input)

The interaction is designed to be simple and intuitive, ensuring ease of use for all users.

Triggering the Workflow

After collecting the required inputs, the system automatically triggers the backend workflow. The user does not need to perform any additional actions.

- The system processes the request in real time
- Identifies nearby shelters
- Initiates communication with shelters automatically

This ensures a seamless and hands-free experience for the donor.

Receiving Outputs

Once a shelter accepts the donation request:

- The donor receives a message containing:
    - Shelter details
    - OTP for verification
- The shelter receives:
    - Donor details

- OTP for confirmation

The user completes the process by entering the OTP through the provided verification mechanism.

Troubleshooting

In case of issues, the following scenarios may occur:

- If no shelters are available within the initial search radius, the system automatically expands the search range
- If all contacted shelters decline the request, the donation may not be completed
- Incorrect input (e.g., wrong zip code or quantity) may lead to inaccurate matching
- Network or call disruptions may require the user to restart the process

Users are encouraged to provide accurate information and ensure stable connectivity for a smooth experience.

10. DEMO INSTRUCTIONS

Demo link

[recording.mp4]

Sample Data

The following sample inputs can be used to demonstrate the system functionality:

- Donation Intent: “I want to donate food”
- Quantity of Food: 50 (or any numeric value via keypad input)
- Zip Code: Valid serviceable zip code (e.g., 600001 or relevant test location)
- Food Type: “Vegetarian meals” or “Packed food”

Additionally, the system uses pre-configured shelter data stored in the database, including:

- Shelter contact numbers
- Geographic location (mapped to zip codes)
- Availability for accepting donations

Execution Steps

1. Initiate the Call

   Dial the configured Twilio phone number to start the demo.
2. Interact with the Voice System

   Follow the voice prompts to:
   - Specify the intent to donate food
   - Enter the quantity using the keypad
   - Enter the zip code via dial pad
   - Provide the type of food through voice input
3. Automated Processing

   The system processes the input and:
   - Identifies nearby shelters based on location
   - Initiates automated calls to shelters sequentially
4. Shelter Response Handling
   - The shelter receives a call with donation details
   - Accepts by pressing the designated key (e.g., 1)

- If rejected, the system proceeds to the next shelter

5. Notification and OTP Sharing

   Once a shelter accepts:

   - Donor receives shelter details and OTP
   - Shelter receives donor details and OTP

6. OTP Verification
   - Both donor and shelter complete OTP verification through the provided mechanism
   - Backend validates the OTP and confirms the transaction
7. Completion

   The donation request is marked as completed and stored in the system.

11. INNOVATION & IMPACT

Business Impact

The proposed solution introduces a structured and efficient approach to food redistribution, addressing a critical gap in the current ecosystem. By automating the process of connecting donors with shelters, the system reduces operational inefficiencies and enables faster decision-making.

- Minimizes food wastage by enabling timely redistribution
- Reduces manual effort and coordination overhead for donors and organizations
- Creates opportunities for NGOs and shelters to receive consistent food support
- Provides a foundation for scalable partnerships with event organizers and institutions
- Enhances overall efficiency in resource utilization within communities

User Impact

The system is designed to deliver a simple and accessible experience for all users, regardless of their technical background.

- Enables donors to contribute effortlessly using a basic phone call
- Reduces time and effort required to identify and contact shelters
- Provides real-time updates and confirmation, improving transparency
- Empowers shelters with timely information to make quick decisions
- Improves access to food for underprivileged communities through faster redistribution

Technical Innovation

The solution demonstrates the effective use of modern cloud technologies combined with a user-centric design approach.

- Implements a voice-first interaction model, eliminating dependency on mobile or web applications
- Integrates conversational AI with serverless backend services for seamless data capture and processing
- Utilizes geospatial intelligence for dynamic and accurate shelter matching
- Employs workflow orchestration to manage complex, multi-step processes with reliability
- Introduces automated, sequential outreach with real-time decision handling

- Incorporates secure OTP-based verification for transaction validation

Scalability Benefits

The system is designed with scalability and future growth in mind, leveraging cloud-native principles.

- Serverless architecture allows automatic scaling based on demand
- Modular design enables easy addition of new features and services
- Can be extended to support multiple regions and larger datasets
- Supports integration with external systems such as NGO platforms, logistics providers, and analytics tools
- Ensures high availability and performance without significant infrastructure investment

12. TECHNICAL DETAILS

API Specifications

The system exposes a set of APIs through Amazon API Gateway to handle communication between external services (Twilio) and backend components.

- Webhook Endpoint (Twilio Integration)
    - Method: POST
    - Purpose: Receives incoming call events and user inputs
    - Processes requests and forwards them to Amazon Lex and backend workflows
- OTP Verification API
    - Method: POST
    - Purpose: Validates OTP entered by donor and shelter
    - Input: OTP, transaction ID or reference ID
    - Output: Verification status (success/failure)
- Donation Processing API (Internal)
    - Triggered via workflow orchestration
    - Handles donation data processing and shelter matching logic

These APIs are secured and designed to handle asynchronous, event-driven interactions.

Data Models

The system uses a NoSQL database (DynamoDB) to store and manage application data. The primary data entities include:

- Donation Request Table
    - Donation ID (Primary Key)
    - Quantity of food
    - Zip code / location
    - Food type
    - Status (Pending, Accepted, Completed)
    - Assigned shelter ID
    - Timestamp
- Shelter Table
    - Shelter ID (Primary Key)
    - Name
    - Contact number

    - Location (coordinates / zip code)
    - Availability status
- OTP / Transaction Data
    - Transaction ID
    - OTP value
    - Expiry time
    - Verification status

The schema is designed for fast access, scalability, and efficient querying.

Security

Security is implemented at multiple levels to ensure safe and reliable operation of the system.

- API Security
    - API Gateway endpoints are configured with controlled access
    - Input validation is enforced to prevent malformed requests
- OTP-Based Verification
    - Ensures authenticity of both donor and shelter
    - Prevents unauthorized or false confirmations
- IAM Roles and Permissions
    - Fine-grained access control between AWS services
    - Ensures least-privilege access for Lambda and other components
- Data Protection
    - Sensitive data is handled securely within managed services
    - Communication between services is protected within the AWS ecosystem

Performance

The system is designed to deliver fast and efficient processing with minimal latency.

- Serverless execution (Lambda) ensures quick response to events
- Step Functions manage workflows efficiently without blocking operations
- DynamoDB provides low-latency read and write operations
- Asynchronous processing enables parallel handling of communication and data operations

These factors ensure that the system responds quickly, which is critical for time-sensitive food redistribution.

Scalability

Scalability is a key strength of the system due to its cloud-native architecture.

- Automatic scaling of Lambda functions based on incoming requests
- API Gateway handles increasing traffic without manual intervention
- DynamoDB supports high throughput with consistent performance
- Step Functions manage increasing workflow executions reliably

The system can scale from a small proof of concept to a large-scale deployment across multiple regions without significant architectural changes.

13. LIMITATIONS

Known Issues

As this solution is developed as a Proof of Concept, certain limitations exist in terms of functionality and robustness:

- The system relies on accurate input from users (e.g., zip code and quantity), and incorrect inputs may lead to suboptimal matching
- Voice recognition accuracy may vary depending on network quality and user input clarity
- Sequential calling of shelters may introduce slight delays if multiple shelters reject the request
- The system currently does not include real-time tracking of food pickup or delivery
- Limited handling of edge cases such as call disconnections or partial input scenarios

Mock Data

The current implementation uses sample or pre-configured data for demonstration purposes:

- Shelter data (location, contact details) is manually populated and may not represent real-world coverage
- Availability status of shelters is assumed and not dynamically updated in real time
- Test scenarios are executed using controlled inputs rather than live production data

This approach was adopted to validate the core functionality and workflow before integrating real-world datasets.

Time Constraints

Due to limited development time, certain enhancements and production-level features were not implemented:

- No dedicated user interface (mobile/web application) beyond a basic OTP verification page
- Limited validation and error-handling mechanisms
- Absence of advanced analytics, monitoring, and reporting features
- No integration with external systems such as NGOs, logistics providers, or government databases

These constraints were considered acceptable for a Proof of Concept, with the primary focus on demonstrating feasibility and core system capabilities.

14. FUTURE ENHANCEMENTS

Additional Features

The current solution can be extended with several features to enhance usability, efficiency, and overall impact:

- Development of a mobile and web application for donors and shelters to provide an alternative to voice interaction
- Implementation of real-time tracking for food pickup and delivery to improve transparency
- Introduction of multi-language support to make the system accessible to a wider audience
- Smart priority-based matching based on urgency, food type, and shelter capacity
- Notification enhancements such as reminders and status updates throughout the donation lifecycle
- Dashboard for monitoring donation activities, trends, and system usage

Integrations

To expand the ecosystem and improve operational effectiveness, the system can be integrated with external platforms and services:

- Integration with NGO and shelter management systems for real-time availability and coordination
- Integration with logistics or delivery services to facilitate transportation of food
- Calendar integration (e.g., scheduling pickups) to enable better planning between donors and shelters
- Integration with government or public welfare databases to extend reach and coverage
- Payment gateway integration for supporting optional donations or operational funding

Production Readiness Steps

To transition the solution from a Proof of Concept to a production-grade system, the following steps are recommended:

- Implementation of robust error handling and retry mechanisms for all workflows
- Enhancement of security measures, including authentication, authorization, and data encryption

- Deployment of monitoring and logging systems for performance tracking and issue detection
- Optimization of database design and indexing for large-scale usage
- Load testing and performance tuning to ensure reliability under high traffic
- Establishment of CI/CD pipelines for continuous deployment and updates
- Expansion of shelter database with verified and real-time data

15. CONCLUSION

Summary of Value

The Food Redistribution System demonstrates how technology can be effectively leveraged to address a real-world social challenge—bridging the gap between surplus food and those in need. By combining voice-based interaction, intelligent location services, and serverless cloud architecture, the solution simplifies the donation process and enables real-time coordination between donors and shelters.

The system eliminates the need for manual effort, reduces delays, and increases the likelihood of successful food redistribution. Its accessible design ensures that anyone with a basic phone can contribute, making it inclusive and practical. At the same time, the use of scalable and managed cloud services ensures that the solution is reliable, efficient, and ready for future expansion.

Final Pitch Message

This project goes beyond being a technical implementation; it represents a meaningful step towards reducing food wastage and improving resource utilization in society. By transforming a donor's intent into immediate action through automation and intelligent workflows, the system creates tangible social impact with minimal user effort.

With further enhancements and real-world deployment, this solution has the potential to evolve into a large-scale platform that supports communities, empowers organizations, and contributes to a more sustainable ecosystem. It showcases not only technical capability but also a problem-solving mindset focused on creating value where it truly matters.

16. APPENDIX

- Docs: API Reference, Tutorials, and Integration | Twilio
- Programmable Voice | Twilio
- AWS Step Functions Documentation
- Amazon Location Service Documentation
- Amazon DynamoDB Documentation
- AWS Lambda Documentation